

Proof. **Step1. Notation.** Let $x_1, \dots, x_n \in \mathbb{R}$ and let $\lambda_1, \dots, \lambda_n \geq 0$ satisfy $\sum_{i=1}^n \lambda_i = 1$. Set

$$X := \sum_{i=1}^n \lambda_i x_i.$$

Step2. Trivial cases. If $\lambda_k = 1$ for some k (hence $\lambda_i = 0$ for $i \neq k$), then $X = x_k$ and the desired inequality reduces to $f(x_k) \leq f(x_k)$, which holds with equality. The same conclusion follows when all points that receive positive weight are equal, because both sides of the inequality are then the common value $f(x)$.

Step3. Base case $n = 2$. For two points x_1, x_2 and a weight $\lambda \in [0, 1]$, convexity of f gives

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

which is Jensen's inequality for $n = 2$.

Step4. Inductive step. Assume the inequality holds for any collection of $n - 1$ points with non-negative weights summing to 1. Without loss of generality suppose $\lambda_n < 1$ (the case $\lambda_n = 1$ is covered by Step2). Write

$$\begin{aligned} X &= \lambda_1 x_1 + \dots + \lambda_{n-1} x_{n-1} + \lambda_n x_n \\ &= (1 - \lambda_n) \left(\sum_{i=1}^{n-1} \frac{\lambda_i}{1 - \lambda_n} x_i \right) + \lambda_n x_n. \end{aligned}$$

Define

$$t := 1 - \lambda_n \in (0, 1], \quad y := \sum_{i=1}^{n-1} \frac{\lambda_i}{1 - \lambda_n} x_i.$$

Because $\sum_{i=1}^n \lambda_i = 1$ we have $\sum_{i=1}^{n-1} \lambda_i = t$; dividing by t yields $\sum_{i=1}^{n-1} \frac{\lambda_i}{t} = 1$. Hence the coefficients $\frac{\lambda_i}{t}$ are non-negative and sum to 1, so y is a convex combination of the first $n - 1$ points.

Step5. Apply convexity to the pair (y, x_n) . Convexity with weight t gives

$$f(ty + (1 - t)x_n) \leq tf(y) + (1 - t)f(x_n)$$

Since $ty + (1 - t)x_n = X$ and $1 - t = \lambda_n$, we obtain

$$f(X) \leq (1 - \lambda_n)f(y) + \lambda_n f(x_n). \quad (5.1)$$

Step6. Use the induction hypothesis on $f(y)$. Applying the induction hypothesis to the $(n - 1)$ -tuple $\{x_1, \dots, x_{n-1}\}$ with the normalized weights $\frac{\lambda_i}{1 - \lambda_n}$ yields

$$f(y) \leq \sum_{i=1}^{n-1} \frac{\lambda_i}{1 - \lambda_n} f(x_i). \quad (6.1)$$

Multiplying (6.1) by $1 - \lambda_n$ and substituting into (5.1) gives

$$\begin{aligned} f(X) &\leq (1 - \lambda_n) \sum_{i=1}^{n-1} \frac{\lambda_i}{1 - \lambda_n} f(x_i) + \lambda_n f(x_n) \\ &= \sum_{i=1}^{n-1} \lambda_i f(x_i) + \lambda_n f(x_n) \\ &= \sum_{i=1}^n \lambda_i f(x_i), \end{aligned}$$

which is Jensen's inequality for n points.

Step7. Equality case. Equality in the binary convexity step (5.1) holds iff either $y = x_n$ or $t \in \{0, 1\}$. Because $t = 1 - \lambda_n$, the latter means $\lambda_n = 0$ or $\lambda_n = 1$. Equality in the induction hypothesis (6.1) holds iff all points with positive normalized weight are identical, i.e. $x_i = x_j$ whenever $\lambda_i > 0$ and $\lambda_j > 0$. Combining these observations, equality in the final inequality occurs precisely when either

- one weight equals 1 (the degenerate case), or
- all points that receive positive weight are equal.

These are exactly the usual equality conditions for Jensen's inequality.

Step8. Conclusion. The base case (Step3) and the inductive step (Steps4–6) establish Jensen's inequality

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

for every integer $n \geq 2$, every convex function f , and every family of non-negative weights summing to 1. The equality cases are described in Step7. $\square$